The classical-to-quantum existence conjecture

Conjecture and scope, 2026-10-27. Every proof of consistency or existence of classical mechanics contains a proof of consistency or existence of quantum mechanics. We interpret this claim constructively: the mechanical data and estimates that establish the classical theory should suffice to construct a compatible quantum theory. The universal claim remains open. For a specified class of smooth scalar potentials, the quantum-to-Newton construction already constructs both dynamics and a quantitative classical limit. Its quantum existence argument uses additional operator theorems; we have not shown that every classical existence argument supplies their required hypotheses.

The immediate question is which classical refinement and stability estimates permit such a construction. Existence of a quantum family for arbitrary supplied positive action, physical necessity of positive action, and uniqueness of a quantum completion are separate claims. No general lifting theorem or independently necessary positive action scale is proved here. The following formulation and comparison use the maintained proofs cited below; they make no literature novelty claim. The subsequent Lipschitz-certificate lift proves the bounded test in §6 for a specified one-dimensional action class, using a named operator-theorem toolkit. The universal claim remains open.

1. The meaning of containment

A proof may contain mathematical information that its author never expresses in quantum terminology. Under the conjecture, a construction of classical motion would also give enough control to construct quantum evolution from the same mechanical data. A direct classical proof that never mentions a Hilbert space therefore does not by itself refute this interpretation. One must determine whether its assumptions and estimates permit the quantum construction.

Literal textual inclusion would be a much stronger requirement on how people write proofs. Conversely, appending an unrelated quantum existence theorem to any classical proof would make containment uninformative. The intended requirement is an explicit construction whose convergence uses the classical data and estimates. Fix a named background analytic toolkit before the test. Every model-dependent hypothesis needed to apply those theorems must be supplied by the classical inputs or proved from them; adding a new dynamical restriction would weaken the containment claim.

The word consistency also needs an identified object. Logical consistency concerns the absence of contradictions in specified axioms. Existence of a mathematical model can establish relative consistency within the mathematical foundation used to construct it. Existence of motion, uniqueness from initial data, continuous dependence, and continuation for all time are different analytic properties. The present programme addresses these analytic constructions and their relation to quantum dynamics. It has not formulated a reduction between formal arithmetical consistency statements for two axiomatic theories.

The quantum target requires a specified noncommutative quantization of the mechanical observables, a self-adjoint Hamiltonian, a strongly continuous unitary evolution, and a relation between that Hamiltonian and the classical action. For the scalar example, the target uses the Schrödinger position and momentum representation, with $Q = q$, $P = -ih\,\partial_q$, the canonical relation $[Q, P] = ihI$ on Schwartz space and its associated Weyl relations, and the Hamiltonian in (2). Other action classes must specify the corresponding quantum structure and correspondence conditions before the lifting test. These are requirements on the constructed output; assuming them as additional physical input would leave the conjecture unproved.

A Hilbert-space representation alone would admit a classical example. For a complete Hamiltonian flow preserving phase volume, the operators $(U(t)f)(z) = f(\Phi_{-t}(z))$ form a strongly continuous unitary group on L^2 of phase space. Stone's theorem gives a self-adjoint generator; multiplying it by any $h > 0$ writes the same group as $e^{-itH_h/h}$. Multiplication observables

commute, and concentrating probability packets recover the classical trajectory. This Koopman representation would satisfy the weaker target without constructing the required noncommutative quantum observables.

A normalizable vacuum, a spectral gap and relativistic field-theory axioms are additional requirements. Free motion on the line has a quantum Hamiltonian and unitary group, although its spectral bottom is not a normalizable vacuum. A mechanical lifting result would not establish the continuum existence or mass gap of four-dimensional Yang–Mills.

Throughout, h has action units and denotes the reduced phase constant. Identification with physical quantum mechanics would set $h = \hbar$; ordinary Planck action is $h_P = 2\pi\hbar$. The time mesh $|\pi|$ is an independent parameter.

2. A testable lifting statement

Fix a class $\mathcal{A}$ of mechanical actions, their state spaces, initial or boundary data, and finite refinement rules. A classical existence certificate means explicit estimates sufficient to construct the classical dynamics. For a polygon proof these may include a common bounded region on each finite time interval, a stability estimate and a force-balance residual tending to zero with the mesh. The admissible certificates must be specified before testing a proposed lift; defining them to include quantum convergence would assume the conclusion.

Constructive lifting conjecture. Fix the action class, certificate schema, quantum target and named background analytic toolkit. One construction procedure for that class takes any admissible mechanical data and classical certificate as input and constructs a compatible quantum family Q_h for $h \in E \subset (0, \infty)$ with $0 \in \overline{E}$. Every model-dependent hypothesis used in the quantum construction must be supplied by those inputs or proved from them. The procedure must identify which certificate estimates control quantum refinement and the recovery of the classical dynamics on the stated preparations and observables. Additional dynamical restrictions cannot be introduced to complete the proof.

The proposed interpretation requires a single construction procedure for the chosen class, rather than a separate unexplained choice for each certificate. A certificate schema must not exclude a counterexample after the test begins. A theorem for one class would establish a bounded lifting result; extending it to every classical existence proof remains the universal conjecture.

Existence at one isolated $h > 0$ cannot itself formulate an $h \to 0$ limit. The lifting statement therefore requires a family indexed by positive values accumulating at zero. Requiring every sufficiently small positive h, or every $h > 0$, strengthens that requirement. The current smooth-potential example constructs the quantum theory for every $h > 0$. This family requirement makes the proposed lifting statement more specific than a bare assertion that some quantum theory exists.

The word compatible requires an observable statement. In the current example, fix z_0, a finite $T > 0$, a time $t \in [0, T]$ and $a \in C_c^\infty(T^*\mathbb{R})$. For the specified coherent preparations and partition vertices $t_j \to t$, the required limit is

$$\left\langle \psi_{h,\pi}(t_j), \mathrm{Op}_h^W(a)\psi_{h,\pi}(t_j) \right\rangle \longrightarrow a(\Phi_t(z_0)), \qquad h, |\pi| \longrightarrow 0. \tag{1}$$

Here z_0 is the initial canonical state, Φ_t the constructed classical flow, $\psi_{h,\pi}(t_j)$ the state after the first j quantum cells, and Op_h^W Weyl quantization. Partitions containing t permit $t_j = t$. The limit holds along every joint schedule with $h, |\pi| \to 0$. A different action class may require different preparations, observables and topologies while retaining its specified noncommutative quantum target. Equation (1) does not require convergence of every quantum state to a single classical point.

3. The smooth-potential instance

The multiscale construction, sections 1–5 (maintained proof read), assumes $m > 0$ and $V \in C^\infty(\mathbb{R})$, with $V \geq 0$ and bounded derivatives of every order at least two. Set $L = \|V''\|_\infty$. Those hypotheses give a globally Lipschitz classical force and define

$$H_h = -\frac{h^2}{2m}\frac{d^2}{dq^2} + V(q), \qquad h > 0. \tag{2}$$

The construction establishes a normalized Euclidean bridge law, the self-adjoint Hamiltonian (2), and arbitrary-partition convergence of real-time kick–drift products at each fixed h. Coherent packet means form a Cauchy family as h and the mesh vanish jointly, without assuming the classical trajectory in advance. Their limit is the unique Newton initial-value solution. The stated quantitative bound is

$$\sup_{0\leq t\leq T} \|z_{h,\pi}(t) - \Phi_t(z_0)\|_\omega \leq G_T\big(C_\delta|\pi| + C_h h\big), \tag{3}$$

with the weighted phase-space norm and constants defined in that proof, equations (17) and (21). They are independent of the number of slices and of $h \in (0, h_0]$ for fixed h_0, T. The sharper $O(h)$ mean term uses the bounded third derivative already included in the hypotheses and cancellation of the centered linear force term; packet widths remain $O(\sqrt{h})$. The groupoid formulation, sections 1–3 (maintained proof read), extends the comparison to smooth Weyl observables and continuous sections of the established quantum/classical field.

The common-hypothesis result has the logical form

$$\mathcal{H} \implies \big[C \text{ and } (\forall h > 0)\, Q_h \text{ and the stated classical limit}\big]. \tag{4}$$

$\mathcal{H}$ denotes the hypotheses just listed, C the classical existence and uniqueness conclusion, and Q_h the specified quantum existence conclusion. Equation (4) records a construction of both theories from common data. It does not establish that the bare assertion C, or every certificate proving C, supplies all information required for Q_h.

The direct polygon proof, sections 1–3 (maintained proof read), constructs classical motion using stability, residual bounds and Gronwall. Its hypotheses admit an arbitrary globally Lipschitz acceleration. The quantum proof additionally specifies a potential, a Hilbert-space realization and operator domains, and uses self-adjointness and product convergence. In particular, positivity of the potential is not needed for the direct classical polygon argument. The quantum-to-classical comparison then uses the same Lipschitz stability to control the packet means. The task is to determine which of these quantum requirements follow from an appropriately specified classical certificate and which require stronger assumptions.

The quartic extension of the packet construction also has a precise boundary. It constructs the classical quartic trajectory from quantum Hamiltonians whose potentials agree with the quartic on a sufficiently large compact interval. Classical energy conservation prevents escape from that interval. It does not prove the joint packet estimate for the uncut quartic quantum Hamiltonian. The separate quartic vacuum construction, sections 2–4 (maintained proof read), establishes its quantum existence and bounded-cylinder reconstruction.

4. The 1998 proposal and phase information

Rivero's 1998 working draft (full-read, sections 1–4, equations (1)–(10)) proposes representing a classical variational distribution through oscillatory functionals. It asks whether convergence of a controlled Feynman functional is equivalent to existence of the proposed classical functional-space distribution. The draft leaves its regulator-control transformation unspecified and permits an arbitrary positive control constant. This is a particular proposed convergence equivalence; the claim about every classical existence proof is broader.

The quartic and central-force tests, section 1 and Theorems 1–3 (maintained proof and corrections read), identify the information that a lifting argument must preserve. At a fixed finite partition, isolated nondegenerate stationary points z_a of S_π give, under the cutoffs and normalization stated there,

$$A_{h,\pi}(O) = \sum_a \frac{O(z_a)}{\sqrt{|\det S''_\pi(z_a)|}} e^{iS_\pi(z_a)/h + i\pi\sigma_a/4} + O_\pi(h). \tag{5}$$

The signature σ_a and the action differences determine relative phases. The classical critical-point distribution, with weights $1/|\det S''_\pi(z_a)|$, retains neither those phases nor a rule for summing amplitudes. The action supplies candidate phases, but convergence of trajectories alone does not bound the oscillatory sum. With several branches, the squared amplitude retains interference terms. The quartic test shows that the unqualified classical-first halved functional can fail to converge even when the ordinary classical initial-value problem is well posed. The draft's single-extremum restriction must be preserved when applying its elementary argument.

The local determinant identity

$$\det S''_\pi = \det S_{yy} \, \det S''_{\text{eff}} \tag{6}$$

holds when a stationary insertion $y = y_*(z)$ has invertible S_{yy}. It relates the classical elimination weight to the square-root weight in stationary phase. A possible lifting mechanism would transport these weights, phases and remainders through refinement. Formula (6) by itself does not control their infinite composition. The $O_\pi(h)$ remainder in (5) is a fixed-partition statement and supplies no uniform mesh limit.

Classical endpoint composition also requires care. Hamilton's principal action composes by stationary elimination on a chosen noncaustic branch. A global minimum requires separate minimizing hypotheses. Quantum propagators compose by integration over intermediate coordinates. These kernel composition laws do not establish that the action is an ordinary groupoid morphism. The corrected groupoid formulation, section 5 (maintained argument read), keeps geometric deformation, time refinement and integration of fluctuations distinct.

5. Completeness and positive action

Classical incompleteness motivates a stronger physical question. For attractive Kepler motion, some initial data reach a collision where the original vector field gives no continuation, whereas the specified Coulomb quantum Hamiltonian has global unitary dynamics for every $h > 0$. The self-sufficiency theorem, Theorem 1(a)–(b) and its dated scope corrections (maintained argument read), establishes this contrast. It establishes neither a contradiction in classical mechanics nor exclusion of all classical completions.

The bounded-cusp theorem, sections 1–4 (maintained proof read), goes further under explicit assumptions. Canonical covariance, weak closure of preparations and reversible compatibility with a force that merges full point states imply a positive covariance action floor. A classical model with a phase-space density ceiling satisfies those same assumptions through an almost-everywhere area-preserving flow. Its preparations give no weight to the exceptional merging trajectories, and it admits no point preparations, so the point-compatibility condition is vacuous in that model. The floor therefore remains insufficient to identify quantum mechanics; record closure and the identification with the phase constant require further arguments.

Existence of a compatible family Q_h for every $h > 0$ leaves all those positive values available and says nothing by itself about the physical admissibility of the zero-action theory. Requiring positive action in nature would need an independently justified condition that excludes the classical alternative. Universality across systems and calibration of one physical value are additional obligations. The constructive lifting conjecture can be investigated before those stronger claims are decided.

6. A bounded test

The test compares the direct polygon certificate with the quantum product argument for a specified action class. Initially fix $m > 0$, $V \geq 0$, the derivative bounds of section 3, the initial data and a finite duration T. Specify the certificate fields as the classical drift and kick rules, their common bounded region, stability and residual estimates, with their domains of validity. Fix the allowed background theorems on closed forms, self-adjoint generators and product limits. Record explicitly which assumptions on V the classical proof uses and which it leaves unused.

Define the quantum finite factors from the same kinetic energy and potential. For fixed $h > 0$, a quantum Cauchy criterion is

$$\|U_{h,\pi}(T)\psi - U_{h,\sigma}(T)\psi\|_2 \leq r_{h,T,\psi}(|\pi|) + r_{h,T,\psi}(|\sigma|), \qquad r_{h,T,\psi}(s) \longrightarrow 0. \tag{7}$$

The subsequent bounded lifting theorem derives the operator-domain and product estimates from the supplied conservative action and its global Lipschitz force bound, using the named Nelson and graph-domain theorems. It actually allows $V \in C^{1,1}$ without a lower bound. For graph-domain vectors it gives an explicit $O(|\pi|)$ fixed-h product estimate, and it uses the classical polygon certificate together with packet-spread bounds to prove an observable limit such as (1). Fixed-h product convergence and the joint classical limit remain different estimates; (7) need not be uniform as $h \to 0$.

That theorem concerns the delimited certificate class, rather than every proof of classical existence. Likewise, nonuniqueness of a quantum boundary condition does not refute a conjecture asking only for existence of a compatible quantum theory.

For singular forces, the cusp-selection calculation gives explicit Gaussian families with the same limiting incoming trajectory and different outgoing branches; their superpositions give mixtures. This reveals preparation information lost by the point-state limit without establishing a universal positive action scale.

7. Consequence for STATE

The universal classical-to-quantum existence conjecture remains open. The subsequent bounded theorem supplies one noncommutative construction from the conservative global-Lipschitz certificate and the named operator toolkit; the cusp examples show why singular continuations require more information. Neither result changes the independent action-necessity and Yang–Mills goals.